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EXPERIMENT-6

Question 1: Find Volume when Mass and Density are Given

Step 1: Problem Statement

Input:

- Mass (m)
- Density (d)

Output:

- Volume (V)

Formula:

Volume = Mass / Density

Step 2: Test Case Table

Test Case	Inputs	Expected Output	Actual Output	Results
1	Mass = 100 kg, Density = 20 kg/m ³	Volume = 5.00 m ³	5.00 m ³	Pass
2	Mass = 50 kg, Density = 10 kg/m ³	Volume = 5.00 m ³	5.00 m ³	Pass
3	Mass = 72 kg, Density = 8 kg/m ³	Volume = 9.00 m ³	9.00 m ³	Pass

Step 3: Algorithm

1. Start
2. Declare variables for mass, density, and volume
3. Enter the values of mass and density
4. Calculate volume using: $\text{Volume} = \text{Mass} / \text{Density}$
5. Display the volume
6. Stop

Step 4: C Program

```
#include <stdio.h>

int main()

{

    float mass, density, volume;

    printf("Enter mass: ");

    scanf("%f", &mass);

    printf("Enter density: ");

    scanf("%f", &density);

    volume = mass / density;

    printf("Volume = %.2f cubic meters\n", volume);

    return 0;

}
```

Step 5: Code Evaluation

Test Case	Actual Output	Results
1	5.00 m ³	Pass
2	5.00 m ³	Pass
3	9.00 m ³	Pass

Step 6: Compilation and Execution

```
(kali㉿kali)-[~]
$ gcc hdoc7.c -o hdoc7

(kali㉿kali)-[~]
$ ./hdoc7
Enter mass: 100
Enter density: 20
Volume = 5.00 cubic meters

(kali㉿kali)-[~]
$ gcc hdoc7.c -o hdoc7

(kali㉿kali)-[~]
$ ./hdoc7
Enter mass: 50
Enter density: 10
Volume = 5.00 cubic meters

(kali㉿kali)-[~]
$ gcc hdoc7.c -o hdoc7

(kali㉿kali)-[~]
$ ./hdoc7
Enter mass: 72
Enter density: 8
Volume = 9.00 cubic meters
```

Question 2: Find Thrust when Pressure and Area are Given

Step 1: Problem Statement

Input:

- Pressure (P)
- Area (A)

Output:

- Thrust (F)

Formula:

Thrust = Pressure $\times$ Area

Step 2: Test Case Table

Test Case	Inputs	Expected Output	Actual Output	Results
1	Pressure = 200 N/m ² , Area = 5 m ²	Thrust = 1000 N	1000N	Pass

2	Pressure = 150 N/m ² , Area = 4 m ²	Thrust = 600 N	600N	Pass
3	Pressure = 80 N/m ² , Area = 10 m ²	Thrust = 800 N	800N	Pass

Step 3: Algorithm

1. Start
2. Declare variables for pressure, area, and thrust
3. Enter pressure and area
4. Calculate thrust using: $\text{Thrust} = \text{Pressure} \times \text{Area}$
5. Display the thrust
6. Stop

Step 4: C Program

```
#include <stdio.h>

int main()
{
    float pressure, area, thrust;

    printf("Enter pressure: ");

    scanf("%f", &pressure);

    printf("Enter area: ");

    scanf("%f", &area);

    thrust = pressure * area;

    printf("Thrust = %.2f Newton\n", thrust);

    return 0;
}
```

Step 5: Code Evaluation

Test Cases	Actual Output	Results
1	1000N	Pass
2	600N	Pass

3	800N	Pass
---	------	------

Step 6: Compilation and Execution

```

(kali㉿kali)-[~]
$ gcc hdoc7ii.c -o hdoc7ii

(kali㉿kali)-[~]
$ ./hdoc7ii
Enter pressure: 200
Enter area: 5
Thrust = 1000.00 Newton

(kali㉿kali)-[~]
$ gcc hdoc7ii.c -o hdoc7ii

(kali㉿kali)-[~]
$ ./hdoc7ii
Enter pressure: 150
Enter area: 4
Thrust = 600.00 Newton

(kali㉿kali)-[~]
$ gcc hdoc7ii.c -o hdoc7ii

(kali㉿kali)-[~]
$ ./hdoc7ii
Enter pressure: 80
Enter area: 10
Thrust = 800.00 Newton

```

Question3: Find Coloumb Force when two charges and distance are given

Step 1: Problem Statement

Input:

- First Charge (q1)
- Second Charge (q2)
- Distance between Charges (r)

Output:

- Coulomb Force (F)

Formula:

$$\text{Force} = (9 \times 10^9 \times q_1 \times q_2) / r^2$$

$$\text{where } k = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$$

Step 2: Test Case Table

Test Case	Inputs	Expected Output	Actual Output	Results
1	q1 = 2×10^{-6} C, q2 = 3×10^{-6} C, r = 2 m	0.0135 N	0.0135 N	Pass
2	q1 = 1×10^{-6} C, q2 = 2×10^{-6} C, r = 1 m	0.0180 N	0.0180 N	Pass
3	q1 = 4×10^{-6} C, q2 = 5×10^{-6} C, r = 2 m	0.0450 N	0.0450 N	Pass

Step 3: Algorithm

1. Start
2. Declare variables for the two charges, distance, constant k, and force
3. Enter the values of the two charges and distance
4. Set the value of k as 9000000000
5. Calculate force using: $\text{Force} = (k \times q1 \times q2) / (r \times r)$
6. Display the force
7. Stop

Step 4: C Program

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    float q1, q2, r, force;
```

```
    float k = 9000000000;
```

```
    printf("Enter first charge: ");
```

```
    scanf("%f", &q1);
```

```
    printf("Enter second charge: ");
```

```
    scanf("%f", &q2);
```

```
    printf("Enter distance: ");
```

```
    scanf("%f", &r);
```

```
    force = (k * q1 * q2) / (r * r);
```

```
    printf("Coulomb Force = %.4f Newton\n", force);
```

```
    return 0;
}
```

Step 5: Code Evaluation

Test Case	Actual Output	Results
1	0.0135 N	Pass
2	0.0180 N	Pass
3	0.0450 N	Pass

Step 6: Compilation and Execution

```
(kali㉿kali)-[~]
└─$ gcc hdoc7iii.c -o hdoc7iii

(kali㉿kali)-[~]
└─$ ./hdoc7iii
Enter first charge: 2e-6
Enter second charge: 3e-6
Enter distance: 2
Coulomb Force = 0.0135 Newton

(kali㉿kali)-[~]
└─$ gcc hdoc7iii.c -o hdoc7iii

(kali㉿kali)-[~]
└─$ ./hdoc7iii
Enter first charge: 1e-6
Enter second charge: 2e-6
Enter distance: 1
Coulomb Force = 0.0180 Newton

(kali㉿kali)-[~]
└─$ gcc hdoc7iii.c -o hdoc7iii

(kali㉿kali)-[~]
└─$ ./hdoc7iii
Enter first charge: 4e-6
Enter second charge: 5e-6
Enter distance: 2
Coulomb Force = 0.0450 Newton
```